BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Measure Theory Crash Course

σ-Algebras, Measures, Outer Measures, Lebesgue Measure, Non Measurable Sets

---

SEPTEMBER 01, 2026

CONDUCTED BY

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

For updates subscribe on @ParthoSutraDhor

What is a Measure?

σ-Algebras

σ-Algebra

Let X be a set. A σ-**algebra** $\mathcal{A}$ on X is a collection of subsets of X such that:

1. $X \in \mathcal{A}$ (the whole set is in the σ-algebra),
2. If $A \in \mathcal{A}$, then $A^c = X \setminus A \in \mathcal{A}$ (closed under complementation),
3. If $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$, then $\cup_{n=1}^{\infty} A_n \in \mathcal{A}$ (closed under countable unions).

Properties of σ-Algebras

Let $\mathcal{A}$ be a σ-algebra on a set X. Then:

1. $\emptyset \in \mathcal{A}$ (the empty set is in the σ-algebra),
2. If $A, B \in \mathcal{A}$, then $A \cap B \in \mathcal{A}$ (closed under finite intersections),
3. If $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$, then $\cap_{n=1}^{\infty} A_n \in \mathcal{A}$ (closed under countable intersections).

Intersection of σ-Algebras

Let $\{\mathcal{A}_i\}_{i\in I}$ be a collection of σ-algebras on a set X. Then the intersection $\cap_{i\in I} \mathcal{A}_i$ is also a σ-algebra on X.

Generated σ-Algebra

Let X be a set and let $\mathcal{C}$ be a collection of subsets of X. The **σ-algebra generated by** $\mathcal{C}$, denoted by $\sigma(\mathcal{C})$, is the smallest σ-algebra on X that contains all the sets in $\mathcal{C}$. Formally, it is defined as:

$$\sigma(\mathcal{C}) = \bigcap\{\mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{C} \subseteq \mathcal{A}\}.$$

Borel σ-Algebra

The **Borel σ-algebra** on the real line $\mathbb{R}$, denoted by $\mathcal{B}(\mathbb{R})$, is the σ-algebra generated by the open intervals in $\mathbb{R}$. That is,

$$\mathcal{B}(\mathbb{R}) = \sigma\left(\{(a, b) : a < b, a, b \in \mathbb{R}\}\right).$$

Size of the Borel σ-Algebra

Show that cardinality of the Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is $\mathfrak{c}$. In other words, show that $\mathcal{B}(\mathbb{R})$ has the same cardinality as the real numbers.

Measure Spaces and Measures

Measurable Space

A **measurable space** is a pair $(X, \mathcal{A})$ where X is a set and $\mathcal{A}$ is a σ-algebra on X. The elements of $\mathcal{A}$ are called **measurable sets**.

Measure

Let $(X, \mathcal{A})$ be a measurable space. A **measure** μ on $(X, \mathcal{A})$ is a function $\mu : \mathcal{A} \to [0, \infty]$ such that:

1. $\mu(\emptyset) = 0$ (the measure of the empty set is zero),
2. (Countable Additivity) For any countable collection of disjoint sets $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

Standard Examples of Measures

Counting Measure

Let X be a set and let $\mathcal{A} = \mathcal{P}(X)$ be the power set of X. Define the **counting measure** $\mu : \mathcal{A} \to [0, \infty]$ by

$$\mu(A) = |A| \quad \text{for all } A \in \mathcal{A},$$

where $|A|$ denotes the cardinality of the set A. Show that μ is a measure on $(X, \mathcal{A})$.

Dirac Measure

Let X be a set and let $x_0 \in X$. Define the **Dirac measure** $\delta_{x_0} : \mathcal{P}(X) \to [0, \infty]$ by

$$\delta_{x_0}(A) = \begin{cases} 1 & \text{if } x_0 \in A, \\ 0 & \text{if } x_0 \notin A, \end{cases}$$

for all $A \subseteq X$. Show that δ_{x_0} is a measure on $(X, \mathcal{P}(X))$.

Borel Measure

Let $\mathcal{B}(\mathbb{R})$ be the Borel σ-algebra on $\mathbb{R}$. Define the **Borel measure** $\mu : \mathcal{B}(\mathbb{R}) \to [0, \infty]$ by

$$\mu((a, b)) = b - a \quad \text{for all } a < b, a, b \in \mathbb{R},$$

and extend it to all Borel sets using the properties of measures. Show that μ is a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Properties of Measures

Let $(X, \mathcal{A}, \mu)$ be a measure space. Then the following properties hold:

1. (Monotonicity) If $A, B \in \mathcal{A}$ and $A \subseteq B$, then $\mu(A) \leq \mu(B)$.
2. (Subadditivity) For any countable collection of sets $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$,
$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$
3. (Continuity from Below) If $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$ is an increasing sequence of sets (i.e., $A_n \subseteq A_{n+1}$ for all n), then
$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n).$$
4. (Continuity from Above) If $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$ is a decreasing sequence of sets (i.e., $A_n \supseteq A_{n+1}$ for all n) and $\mu(A_1) < \infty$, then
$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n).$$

Borel Measure of $\mathbb{Q}$

Show that the Borel measure of the set of rational numbers $\mathbb{Q}$ is zero, i.e., $\mu(\mathbb{Q}) = 0$.

Borel Measure of a Singleton

Show that the Borel measure of a singleton set $\{x\}$, where $x \in \mathbb{R}$, is zero, i.e., $\mu(\{x\}) = 0$.

Borel Measure of a countable set

Show that the Borel measure of any countable set $A \subseteq \mathbb{R}$ is zero, i.e., $\mu(A) = 0$.

Outer Measures

Outer Measure

An **outer measure** on a set X is a function $\mu^* : \mathcal{P}(X) \to [0, \infty]$ such that:

1. $\mu^*(\emptyset) = 0$ (the outer measure of the empty set is zero),
2. (Monotonicity) If $A, B \subseteq X$ and $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$,
3. (Countable Subadditivity) For any countable collection of sets $\{A_n\}_{n=1}^{\infty} \subseteq X$,

$$\mu^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n).$$

Length of an Interval in $\mathbb{R}$

The **length** of an interval $I = (a, b)$, where $a < b$, is defined as $\ell(I) = b - a$. For other types of intervals, the length is defined similarly:

- For a closed interval $[a, b]$, $\ell([a, b]) = b - a$.
- For a half-open interval $(a, b]$ or $[a, b)$, $\ell((a, b]) = \ell([a, b)) = b - a$.

Volume of a Rectangle in $\mathbb{R}^n$

Let $R = \Pi_{i=1}^n [a_i, b_i]$ be a rectangle in $\mathbb{R}^n$, where $a_i < b_i$ for all $i = 1, 2, \ldots, n$. The **volume** of the rectangle R is defined as

$$\text{Vol}(R) = \prod_{i=1}^{n} (b_i - a_i).$$

Lebesgue outer measure in $\mathbb{R}$

The **Lebesgue outer measure** on $\mathbb{R}$ is the function $\mu^* : \mathcal{P}(\mathbb{R}) \to [0, \infty]$ defined by

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \ell(I_i) : A \subseteq \bigcup_{i=1}^{\infty} I_i \right\}$$

for all $A \subseteq \mathbb{R}$, where the infimum is taken over all countable collections of intervals $\{I_i\}_{i=1}^{\infty}$ that cover A.

Lebesgue outer measure in $\mathbb{R}^n$

The **Lebesgue outer measure** on $\mathbb{R}^n$ is the function $\mu^* : \mathcal{P}(\mathbb{R}^n) \to [0, \infty]$ defined by

$$\mu^*(A) = \inf\left\{\sum_{i=1}^{\infty} \mathsf{Vol}(R_i) : A \subseteq \bigcup_{i=1}^{\infty} R_i\right\}$$

for all $A \subseteq \mathbb{R}^n$, where the infimum is taken over all countable collections of rectangles $\{R_i\}_{i=1}^{\infty}$ that cover A.

Outer measure of an interval

Let $I = (a, b)$ be an interval in $\mathbb{R}$. Then the Lebesgue outer measure of I is equal to its length, i.e.,

$$\mu^*(I) = \ell(I) = b - a.$$

Outer measure of a rectangle

Let $R = \prod_{i=1}^{n}[a_i, b_i]$ be a rectangle in $\mathbb{R}^n$. Then the Lebesgue outer measure of R is equal to its volume, i.e.,

$$\mu^*(R) = \mathsf{Vol}(R) = \prod_{i=1}^{n}(b_i - a_i).$$

Translation invariance of outer measure

Let $A \subseteq \mathbb{R}$ and $c \in \mathbb{R}$. Then the Lebesgue outer measure of the translated set $A + c = \{x + c : x \in A\}$ is equal to the Lebesgue outer measure of A, i.e.,

$$\mu^*(A + c) = \mu^*(A).$$

Outer measure of a countable set

Let $A \subseteq \mathbb{R}$ be a countable set. Then the Lebesgue outer measure of A is zero, i.e.,

$$\mu^*(A) = 0.$$

Carathéodory's Criterion for Measurability

Carathéodory Measurable Set

A set $E \subseteq \mathbb{R}$ is said to be **Carathéodory measurable** (or simply **measurable**) with respect to the Lebesgue outer measure μ^* if for every set $A \subseteq \mathbb{R}$,

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c).$$

Carathéodory's theorem

The collection of all Carathéodory measurable sets forms a σ-algebra, and the restriction of the Lebesgue outer measure μ^* to this σ-algebra is a measure, called the **Lebesgue measure**.

Lebesgue Measure

Lebesgue Measure

The **Lebesgue measure** μ on $\mathbb{R}$ is the measure defined on the σ-algebra of Carathéodory measurable sets, which is a subset of the Borel σ-algebra. For any Carathéodory measurable set $E \subseteq \mathbb{R}$, the Lebesgue measure is given by

$$\mu(E) = \mu^*(E),$$

where μ^* is the Lebesgue outer measure.

Translation Invariance of Lebesgue Measure

An important geometric property of Lebesgue measure is its translational in-variance. The Lebesgue measure μ is said to be **translation invariant** if for any measurable set $E \subseteq \mathbb{R}$ and any real number $c \in \mathbb{R}$, the translated set $E + c = \{x + c : x \in E\}$ is also measurable and

$$\mu(E + c) = \mu(E).$$

Null Set

A set $N \subseteq \mathbb{R}$ is called a **null set** (or a set of measure zero) if its Lebesgue measure is zero, i.e.,

$$\mu(N) = 0.$$

Complete Measure Space

A measure space $(X, \mathcal{A}, \mu)$ is said to be **complete** if every subset of a null set is measurable. That is, if $N \in \mathcal{A}$ with $\mu(N) = 0$ and $S \subseteq N$, then $S \in \mathcal{A}$ and $\mu(S) = 0$.

Completeness of Lebesgue Measure

The Lebesgue measure μ on $\mathbb{R}$ is a complete measure. That is, if $N \subseteq \mathbb{R}$ is a null set (i.e., $\mu(N) = 0$) and $S \subseteq N$, then S is Lebesgue measurable and $\mu(S) = 0$.

Lebesgue Measure of a countable set

Let $A \subseteq \mathbb{R}$ be a countable set. Then the Lebesgue measure of A is zero, i.e.,

$$\mu(A) = 0.$$

Is there any uncountable set of measure zero?

Can you give an example of an uncountable set of measure zero?

Cantor Set definition

The Cantor set C is constructed by iteratively removing the open middle third intervals from the closed interval $[0, 1]$. Formally, we define the Cantor set as follows:

1. Start with the closed interval $C_0 = [0, 1]$.
2. Remove the open middle third interval $(\frac{1}{3}, \frac{2}{3})$ to obtain $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$.
3. In each subsequent step, remove the open middle third intervals from each remaining closed interval. For example, in the second step, remove $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$ from C_1 to obtain $C_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$, and so on.
4. The Cantor set is defined as the intersection of all these sets:

$$C = \bigcap_{n=0}^{\infty} C_n.$$

Cantor set has measure zero

Show that the Cantor set C has Lebesgue measure zero, i.e., $\mu(C) = 0$.

Cantor set is uncountable

Show that the Cantor set C is uncountable.

Size of the Lebesgue σ-Algebra

Show that the cardinality of the Lebesgue σ-algebra is $2^{\mathfrak{c}}$. In other words, show that the Lebesgue σ-algebra has the same cardinality as the power set of the real numbers.

Almost Everywhere notation

Almost Everywhere

A property $P(x)$ is said to hold **almost everywhere** (a.e.) on a measure space $(X, \mathcal{A}, \mu)$ if the set of points where $P(x)$ does not hold has measure zero. Formally, $P(x)$ holds almost everywhere if

$$\mu(\{x \in X : P(x) \text{ does not hold}\}) = 0.$$

Is all subsets of $\mathbb{R}^n$ Lebesgue measurable?

Non measurable sets

Give an example of a non-measurable set in $\mathbb{R}$. (Hint: Consider the Vitali set.)

Some Important Results and Notes

Why Countable Additivity is Important?

Explain why countable additivity is an important property for a measure. What would go wrong if we allow uncountable additivity instead? Also why finite additivity is not enough to define a measure?

Is all subsets of $\mathbb{R}^n$ Lebesgue measurable?

It is not possible to define the Lebesgue measure of all subsets of $\mathbb{R}^n$. Hausdorff (1914) showed that for any dimension $n \geq 1$, there is no countably additive measure defined on all subsets of $\mathbb{R}^n$ that is translation invariant and assigns measure 1 to the unit cube. In particular, there exist non-measurable sets in $\mathbb{R}^n$.

He further showed that if $n \geq 3$, there is no such finitely additive measure. This result is dramatized by the Banach-Tarski paradox, which states that a solid ball in 3-dimensional space can be split into a finite number of pieces and reassembled into two solid balls identical to the original. This paradox relies on the existence of non-measurable sets.

Thank You!

We'd love your questions and feedback.

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

@ParthoSutraDhor

(Lectures, walkthroughs, and course updates)

Scan for the channel

References

[1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.

[2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.